

Merging Data Frames in R Programming

Name: Savitha R

Reg.no: 192421368

Merging data frames is an essential task in R programming when working with multiple datasets.

The `merge()` function and the `dplyr` package are commonly used for combining datasets.

1. Using merge():

The `merge()` function combines two data frames by common columns or row names.

Syntax:

```
merge(x, y, by = "column_name")
```

Example:

```
df1 <- data.frame(ID = c(1,2,3), Name =  
c("A","B","C"))
```

```
df2 <- data.frame(ID = c(1,2,3), Score = c(90,85,88))
```

```
merged_df <- merge(df1, df2, by = "ID")
```

2. Types of Merge:

- Inner Join: `merge(df1, df2, by = "ID")`
- Left Join: `merge(df1, df2, by = "ID", all.x = TRUE)`
- Right Join: `merge(df1, df2, by = "ID", all.y = TRUE)`
- Full Join: `merge(df1, df2, by = "ID", all = TRUE)`

3. Using dplyr for merging:

```
library(dplyr)
```

```
left_join(df1, df2, by = "ID")
```

```
right_join(df1, df2, by = "ID")
```

```
inner_join(df1, df2, by = "ID")
```

```
full_join(df1, df2, by = "ID")
```

4. Key Points:

- Ensure column names match before merging.
- Check for duplicate keys to avoid unexpected row expansion.
- Merging helps combine attributes from multiple datasets for analysis.

Topic 1: Data Reshaping in R

Q1. Create a data frame and display its structure.

Answer:

```
student <- data.frame(  
  Name = c("Asha", "Ravi", "Kiran"),  
  Marks = c(85, 90, 78)  
)  
str(student)
```

Output:

'data.frame': 3 obs. of 2 variables

```
>  
> student <- data.frame(  
+   ID = c(1, 2, 3),  
+   Name = c("Ravi", "Priya", "Arun"),  
+   Marks = c(85, 90, 78)  
+ )  
>  
> print(student)  
  ID Name Marks  
1  1  Ravi    85  
2  2 Priya    90  
3  3  Arun    78
```

Q2. Rename the column "Marks" to "Score".

Answer:

```
student <- data.frame(  
  Name = c("Asha", "Ravi"),  
  Marks = c(85, 90)
```

)

names(student)[2] <- "Score"

student

Output

Name Score

1 Asha 85

2 Ravi 90

```
> student <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> names(student)[2] <- "Score"
>
> student
  Name Score
1 Asha   85
2 Ravi   90
```

Q3. Add a new column called "Grade".

Answer:

student <- data.frame(

Name=c("Asha","Ravi"),

Marks=c(85,90)

)

student\$Grade <- c("A","A")

student

```
> student <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> student$Grade <- c("A", "A")
>
> student
  Name Marks Grade
1 Asha   85     A
2 Ravi   90     A
```

Q4. Delete the Grade column.

Answer

student\$Grade <- NULL

student

```
> student$Grade <- NULL
> student
  Name Marks
1 Asha    85
2 Ravi    90
```

Q5. Change the order of columns.

Answer

student <- student[, c("Marks","Name")]

student

```
>
> student <- student[, c("Marks","Name")]
> student
  Marks Name
1    85 Asha
2    90 Ravi
```

Topic 2: Melting Data

Q6. Convert a wide-format data frame into long format using melt().

Answer

library(reshape2)

data <- data.frame(

 Name=c("A","B"),

 Math=c(80,85),

 Science=c(90,88)

)

melt(data, id.vars="Name")

Output

Name variable value

A Math 80

B Math 85

A Science 90

B Science 88

```

>
> library(reshape2)
>
> data <- data.frame(
+   Name=c("A","B"),
+   Math=c(80,85),
+   Science=c(90,88)
+ )
>
> melt(data, id.vars="Name")
  Name variable value
1    A      Math    80
2    B      Math    85
3    A   Science    90
4    B   Science    88

```

Q7. Melt multiple subject columns.

Answer

```
library(reshape2)
```

```
marks <- data.frame(
```

```
  Student=c("S1","S2"),
```

```
  English=c(75,80),
```

```
  Math=c(82,90),
```

```
  Science=c(88,91)
```

```
)
```

```
melt(marks,id.vars="Student")
```

```

> library(reshape2)
>
> marks <- data.frame(
+   Student=c("S1","S2"),
+   English=c(75,80),
+   Math=c(82,90),
+   Science=c(88,91)
+ )
>
> melt(marks, id.vars="Student")
  student variable value
1      S1   English    75
2      S2   English    80
3      S1     Math    82
4      S2     Math    90
5      S1  Science    88
6      S2  Science    91

```

Q8. Use variable.name and value.name in melt().

Answer

```
library(reshape2)

melt(data,

id.vars="Name",

variable.name="Subject",

value.name="Marks")
```

```
> library(reshape2)
>
> melt(data,
+       id.vars="Name",
+       variable.name="Subject",
+       value.name="Marks")
  Name Subject Marks
1    A    Math    80
2    B    Math    85
3    A Science    90
4    B Science    88
>
```

Q9. Melt employee salary data.

Answer

```
library(reshape2)

emp <- data.frame(

Employee=c("E1","E2"),

Jan=c(25000,28000),

Feb=c(26000,28500)

)

melt(emp,id.vars="Employee")
```

```
>
> library(reshape2)
>
> emp <- data.frame(
+   Employee=c("E1","E2"),
+   Jan=c(25000,28000),
+   Feb=c(26000,28500)
+ )
>
> melt(emp, id.vars="Employee")
  Employee variable value
1       E1      Jan 25000
2       E2      Jan 28000
3       E1      Feb 26000
4       E2      Feb 28500
```

Q10. Melt sales data.

Answer

```
library(reshape2)

sales <- data.frame(
  Product=c("Pen","Book"),
  Jan=c(50,80),
  Feb=c(70,90)
)

melt(sales,id.vars="Product")
```

```
>
> library(reshape2)
>
> sales <- data.frame(
+   Product=c("Pen","Book"),
+   Jan=c(50,80),
+   Feb=c(70,90)
+ )
>
> melt(sales, id.vars="Product")
  Product variable value
1     Pen      Jan    50
2     Book      Jan    80
3     Pen      Feb    70
4     Book      Feb    90
```

Topic 3: Casting Data

Q11. Convert melted data back into wide format.

Answer

```
library(reshape2)

longdata <- melt(data,id.vars="Name")

dcast(longdata,
      Name~variable)
```

```
>
> library(reshape2)
>
> longdata <- melt(data, id.vars="Name")
>
> dcast(longdata, Name~variable)
  Name Math Science
1    A   80     90
2    B   85     88
```

Q12. Create subjects as columns.

Answer

```
dcast(longdata,  
      Name~variable,  
      value.var="value")
```

Q13. Find average marks using dcast().

Answer

```
dcast(longdata,  
      Name~variable,  
      mean)
```

```
> dcast(longdata,  
+       Name~variable,  
+       value.var="value")  
   Name Math Science  
1     A    80      90  
2     B    85      88
```

Q14. Count observations using dcast().

Answer

```
dcast(longdata,  
      Name~variable,  
      length)
```

```
> dcast(longdata,  
+       Name~variable,  
+       length)  
   Name Math Science  
1     A     1      1  
2     B     1      1
```

Q15. Reshape employee data using dcast().

Answer

```
emp.long <- melt(emp,id.vars="Employee")
```



```
dcast(emp.long,  
Employee~variable)
```

```
> library(reshape2)  
>  
> emp.long <- melt(emp, id.vars="Employee")  
>  
> dcast(emp.long,  
+       Employee~variable)  
Employee Jan Feb  
1      E1 25000 26000  
2      E2 28000 28500
```

Topic 4: Merging Data Frames

Q16. Merge two data frames using a common column.

Answer

```
student <- data.frame(  
ID=c(1,2),  
Name=c("Asha","Ravi")  
)  
  
marks <- data.frame(  
ID=c(1,2),  
Marks=c(90,85)  
)  
  
merge(student,marks,by="ID")
```

```
>  
> student <- data.frame(  
+   ID=c(1,2),  
+   Name=c("Asha","Ravi")  
+ )  
>  
> marks <- data.frame(  
+   ID=c(1,2),  
+   Marks=c(90,85)  
+ )  
>  
> merge(student, marks, by="ID")  
  ID Name Marks  
1  1 Asha   90  
2  2 Ravi   85
```

Q17. Perform a left join.

Answer

```
merge(student,  
marks,  
by="ID",  
all.x=TRUE)
```

```
> merge(student,  
+       marks,  
+       by="ID",  
+       all.x=TRUE)  
  ID Name Marks  
1  1  Asha   90  
2  2   Ravi   85
```

Q18. Perform a right join.

Answer

```
merge(student,  
marks,  
by="ID",  
all.y=TRUE)
```

```
> merge(student,  
+       marks,  
+       by="ID",  
+       all.y=TRUE)  
  ID Name Marks  
1  1  Asha   90  
2  2   Ravi   85
```

Q19. Perform a full outer join.

Answer

```
merge(student,  
marks,  
by="ID",
```

all=TRUE)

```
<
> merge(student,
+       marks,
+       by="ID",
+       all=TRUE)
  ID Name Marks
1  1  Asha   90
2  2   Ravi   85
```

Q20. Merge two data frames with different column names.

Answer

student <- data.frame(

SID=c(1,2),

Name=c("A","B"))

)

marks <- data.frame(

ID=c(1,2),

Marks=c(90,80)

)

merge(student,

marks,

by.x="SID",

by.y="ID")

```
> student <- data.frame(
+   SID=c(1,2),
+   Name=c("A","B"))
>
> marks <- data.frame(
+   ID=c(1,2),
+   Marks=c(90,80))
>
> merge(student,
+       marks,
+       by.x="SID",
+       by.y="ID")
  SID Name Marks
1   1    A    90
2   2    B    80
```

Additional Practice Questions

Q21. Create a data frame with student names and marks and reshape it into long format.

Answer

```
library(reshape2)

df <- data.frame(
  Name=c("A","B"),
  Math=c(90,85),
  Science=c(80,88)
)

melt(df,id.vars="Name")
```

```
> library(reshape2)
>
> df <- data.frame(
+   Name=c("A","B"),
+   Math=c(90,85),
+   Science=c(80,88)
+ )
>
> melt(df, id.vars="Name")
  Name variable value
1    A      Math    90
2    B      Math    85
3    A  Science    80
4    B  Science    88
```

Q22. Convert long-format data back into wide format.

Answer

```
long <- melt(df,id.vars="Name")

dcast(long,Name~variable)
```

```
> long <- melt(df, id.vars="Name")
>
> dcast(long, Name~variable)
  Name Math Science
1    A    90     80
2    B    85     88
```

Q23. Merge employee and department data frames.

Answer

```
emp <- data.frame(
  ID=c(101,102),
  Name=c("John","Mary")
)

dept <- data.frame(
  ID=c(101,102),
  Department=c("HR","IT")
)

merge(emp,dept,by="ID")
```

```
.
> emp <- data.frame(
+   ID=c(101,102),
+   Name=c("John","Mary")
+ )
>
> dept <- data.frame(
+   ID=c(101,102),
+   Department=c("HR","IT")
+ )
>
> merge(emp, dept, by="ID")
  ID Name Department
1 101 John          HR
2 102 Mary          IT
```

Q24. Display all records after merging.

Answer

```
merge(emp,dept,
  by="ID",
  all=TRUE)
```

```

> merge(emp,
+       dept,
+       by="ID",
+       all=TRUE)
  ID Name Department
1 101 John          HR
2 102 Mary          IT
> merge(emp, dept, by="ID")
  ID Name Department
1 101 John          HR
2 102 Mary          IT

```

Q25. Reshape monthly sales data from wide to long format.

Answer

```
library(reshape2)
```

```
sales <- data.frame(
```

```
  Product=c("Pen","Book"),
```

```
  Jan=c(100,200),
```

```
  Feb=c(150,250)
```

```
)
```

```
melt(sales,id.vars="Product")
```

```

> library(reshape2)
>
> sales <- data.frame(
+   Product=c("Pen","Book"),
+   Jan=c(100,200),
+   Feb=c(150,250)
+ )
>
> melt(sales, id.vars="Product")
  Product variable value
1     Pen      Jan   100
2    Book      Jan   200
3     Pen      Feb   150
4    Book      Feb   250

```